

#### PROBLEM STATEMENT

TERADYNE's current **ARO projection** process faces challenges in providing actionable insights, often leading to **manual interventions** by the Revenue & Operations group. Variations between projected and actual AROs contribute to missed **On-Time Delivery (OTD)** targets. The company measures OTD and **customer satisfaction** using 3 key metrics:

- **OTD to Factory Promise Date**
- **OTD to Advertised ARO**
- **OTD to Customer Request Date**

Another ongoing challenge is **optimizing inventory levels (DIO)**, as both insufficient and excessive inventory directly impact delivery **lead times**, production efficiency, and carrying costs.

#### BUSINESS GOALS

##### DESCRIPTIVE ANALYSIS

- Evaluate **Projected ARO** accuracy vs Actual ARO
- Analyze **OTD performance** vs Promised Date
- The **customer satisfaction** metrics include:
  - OTD vs Advertised ARO
  - OTD vs Customer Request Date



##### PRESCRIPTIVE ANALYSIS

- Explore machine learning techniques to improve **ARO forecasting**
- Recommend how **inventory levels** could be adjusted to achieve 100% OTD at both 16 and 10 weeks

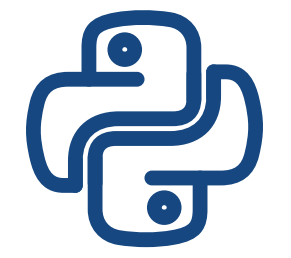
#### TOOLS & METHODS



LAKEHOUSE



POWER BI



PYTHON



Standardize  
Date Format

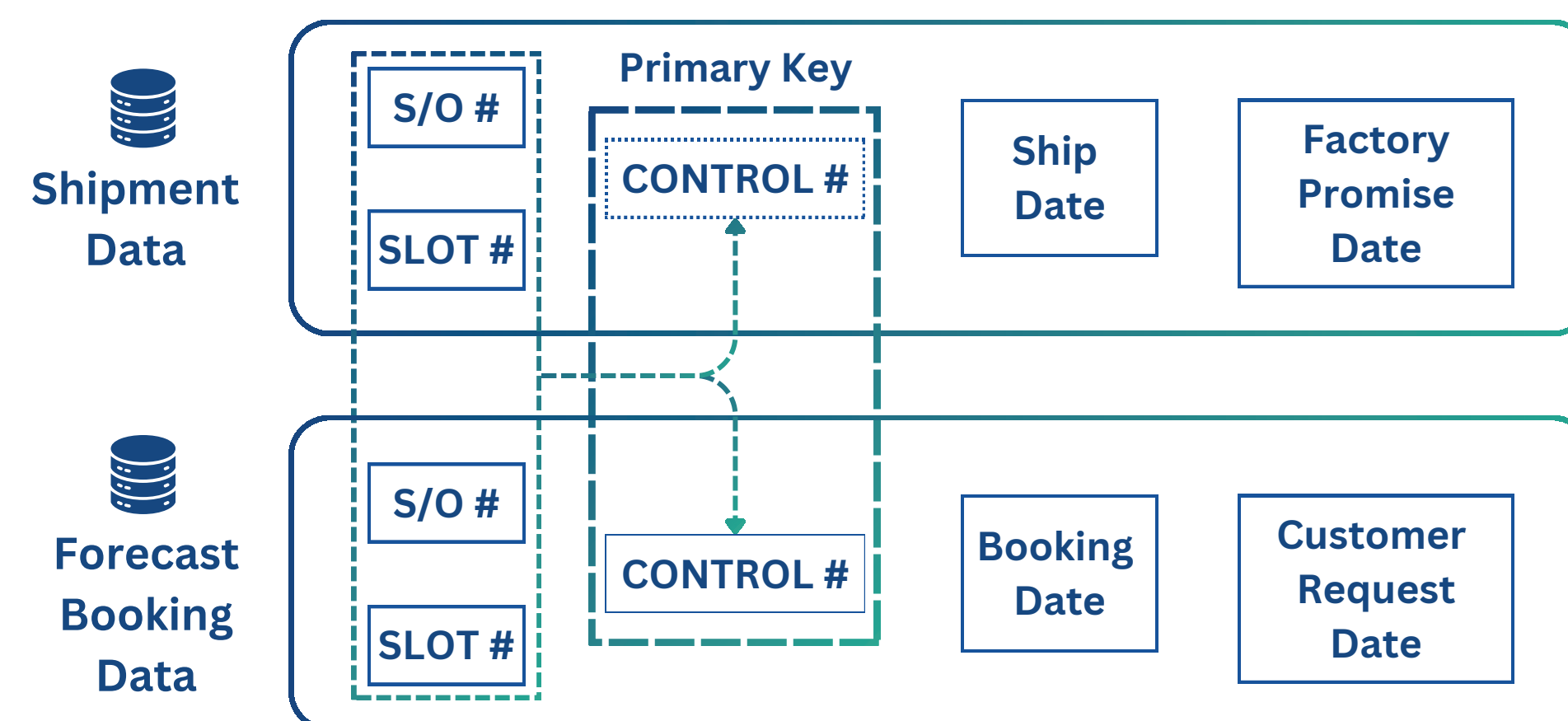


Cycle Time  
Calculation

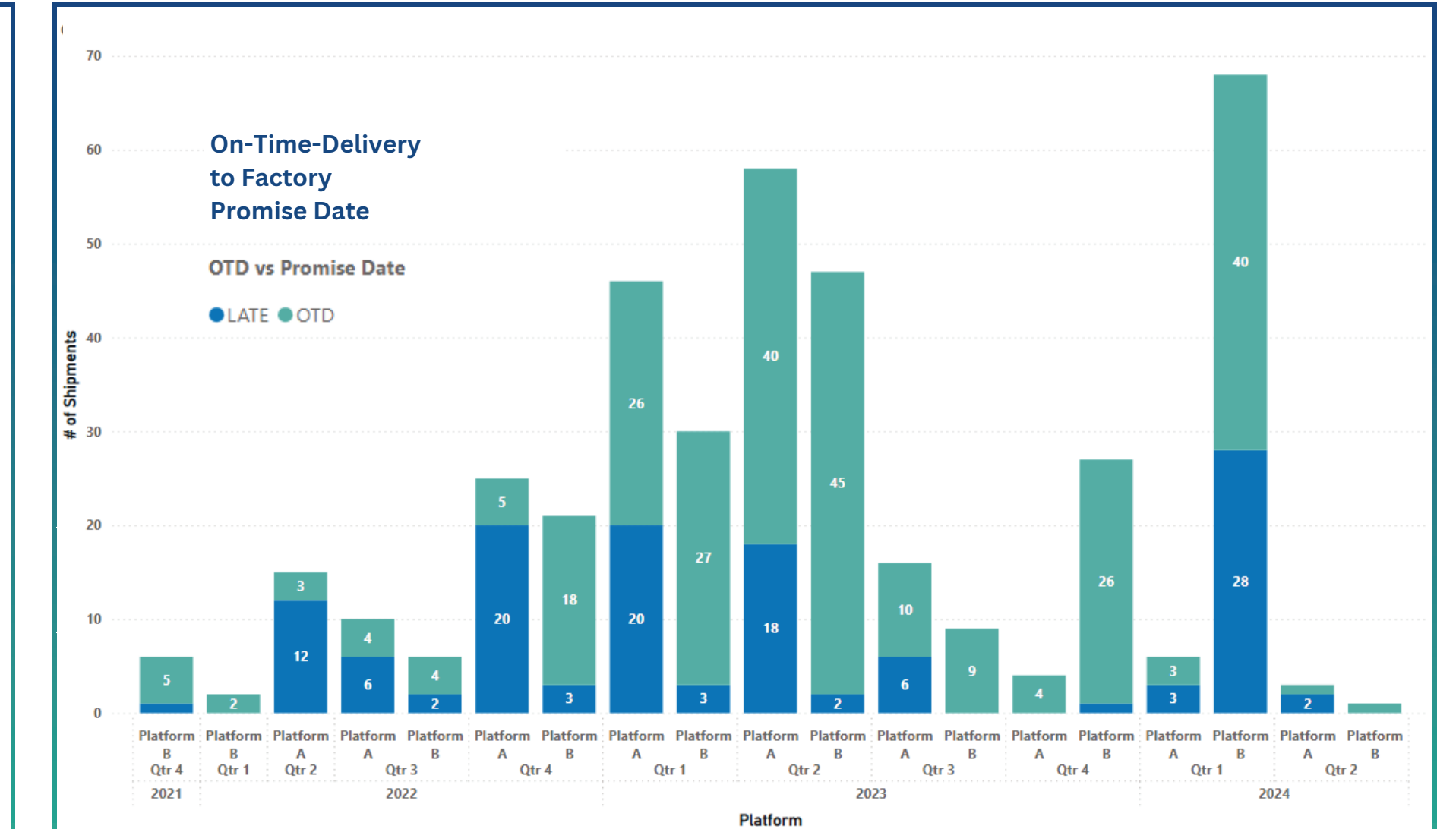
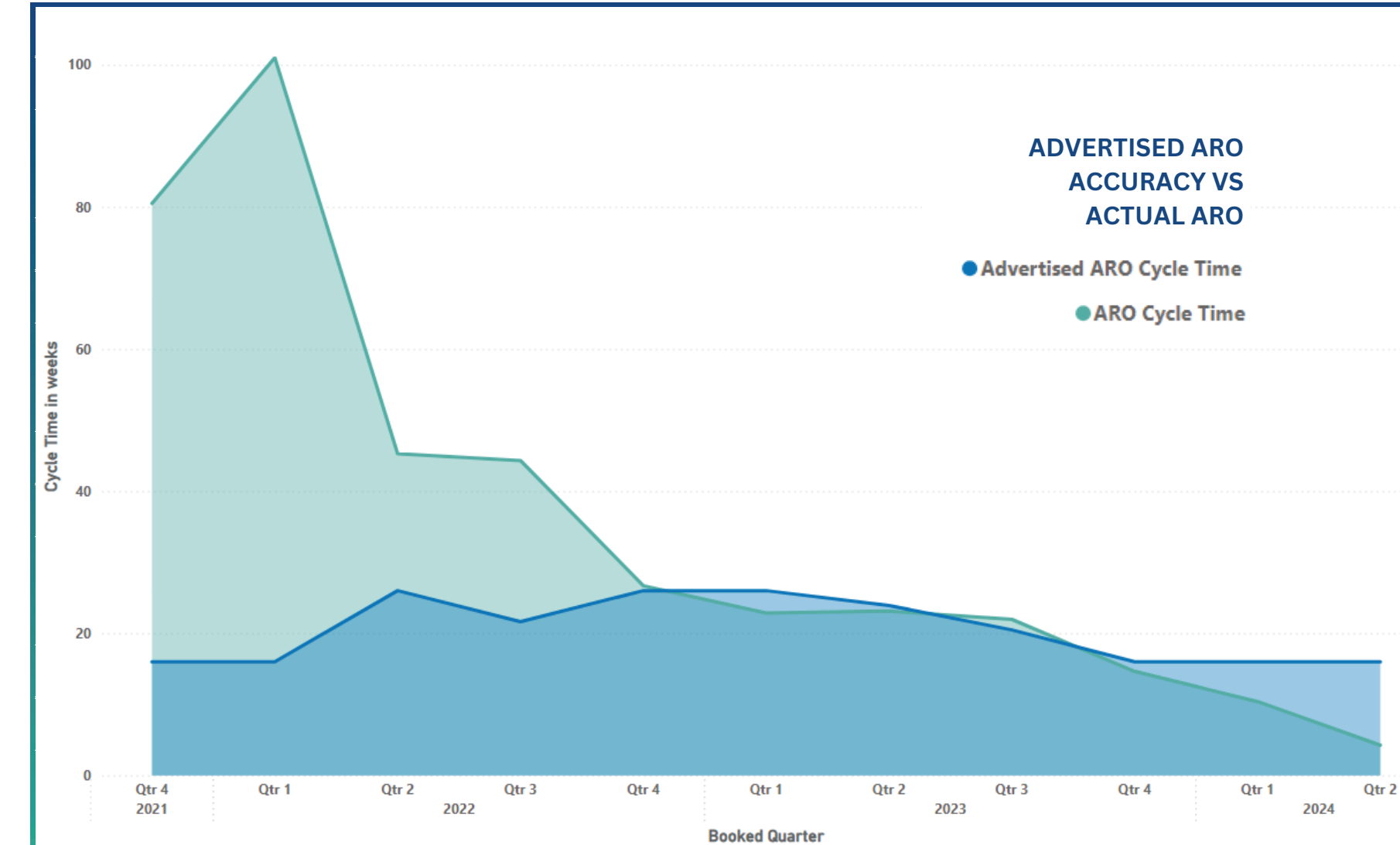


Determine  
On-Time Delivery

#### Data Engineering



#### Analysis & Insights



#### Future Scope



Analyze OTD  
for Other  
Platforms



Forecasting  
Model Adv  
ARO



Optimize  
Inventory  
Levels (DIO)



Benchmark  
Analysis

